Isha Rajwar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.28
Spring Dale College	Class XII (CISCE)	2020	88.20%
Spring Dale College	Class X (CISCE)	2018	93.33%

PROJECTS

Real-Time Radar Deinterleaving on FPGA-ARM SoC Github

- Worked on a **DRDO-associated radar signal processing project**, developing an incremental deinterleaving pipeline using K-Means clustering, Hierarchical Agglomerative Clustering (HAC), and PRI-based validation to dynamically form, merge, and filter emitter clusters from streaming PDW data.
- Built a real-time Pulse Descriptor Word (PDW) pipeline using PS-PL hardware/software partitioning on a ZCU102 ARM-FPGA SoC; developed custom HLS accelerators for clustering and PRI filtering using memory-mapped interfaces.
- Achieved approximately **25× on-board speedup** on selected datasets, even without PS-side optimization, while optimizing accelerator latency and FPGA resource utilization.
- Tech Stack: C/C++, Vitis HLS, ZCU102, FPGA

ML-Based Auto-Tuned Matrix Multiplication Github

- Implemented and benchmarked naive, loop-reordered, cache-blocked, OpenMP, blocked-OpenMP and ARM NEON SIMD kernels to study cache locality, parallelism and vectorization.
- Built an ML-based auto-tuner using Random Forest and Extra Trees; Top-3 hybrid selection reduced tuning candidates by **~95%** while maintaining near-oracle performance.
- Tech Stack : C++, ARM NEON, OpenMP, SIMD

Concurrent Banking System Github

- Built a multithreaded banking simulator supporting concurrent deposits and blocking withdrawals over shared accounts.
- Implemented synchronization and priority handling using mutexes, condition variables and atomics; evaluated throughput under varying thread counts and contention.
- Tech Stack : C++, pthreads, Mutexes, Condition Variables, Atomics

Distributed Minimum Spanning Tree Algorithms Github

- Implemented GHS, Controlled GHS and Borvka-style distributed MST algorithms in C++ using MPI-based inter-process communication.
- Designed distributed graph execution with fragment coordination, message passing, BFS-based topology construction and minimum-edge selection across varied network topologies.
- Benchmarked and analyzed execution time, communication overhead and scalability for the distributed MST implementations.
- Tech Stack: C++, MPI, Python, Jupyter Notebook

FedLSM++ -- Uncertainty-Aware Federated Learning Github

- Extended FedLSM for Label Set Mismatch by replacing entropy-based uncertainty with a novel Monte Carlo variance-ratio estimator over stochastic augmented views and incorporating MC Dropout for epistemic uncertainty.
- Improved F1 from **0.7130 to 0.7336** on ISIC2018 and **0.3546 to 0.3848** on CIFAR-10 using uncertainty-aware pseudo-label selection, while achieving the gains with fewer training rounds.
- Tech Stack : Python, PyTorch, NumPy, Scikit-learn, MC Dropout, Federated Learning.

SCHOLASTIC ACHIEVEMENTS

AKTU University Rank 6: Secured 6th rank across **400+** AKTU-affiliated engineering colleges during B.Tech.

HiPC Student Research Symposium 2025: Research work titled *"Density-Hierarchical Clustering with PRI Transform for Robust Radar Signal Deinterleaving"* accepted at **HiPC SRS 2025**. Publication

Smart India Hackathon (SIH) 2026 -- Internal Hackathon Qualifier : Qualified through the IIT Hyderabad Internal Hackathon with **ParaFleet**, an Edge-AI based distributed AMR fleet coordination solution for a BEL problem statement.

SKILLS

Languages: C, C++, Python, Shell/Bash

FPGA & Hardware: FPGA, ZCU102, Vitis HLS, Vivado, ARM, Memory-Mapped I/O, Hardware-Software Co-design

Parallel Programming: Linux, pthreads, OpenMP, SIMD, ARM NEON, Mutexes, Condition Variables, Atomics

Machine Learning: PyTorch, Scikit-learn, Federated Learning, Uncertainty Estimation

Tools: Git, GCC/G++, Clang, Jupyter/Colab, Python Automation

POSITIONS OF RESPONSIBILITY

Teaching Assistant, IIT Hyderabad: Parallel & Concurrent Programming, Operating Systems II and Concurrency Control in Transactional Systems.

RELEVANT COURSES

Parallel & Concurrent Programming **|** Advanced Data Structures & Algorithms **|** Foundations of Machine Learning **|** Distributed Computing **|** Concurrency Control in Transactional Systems **|** Deep Learning **|** Operating Systems